

ECSE-4040

Digital Electronics

Lecture 23

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Lecture Outline

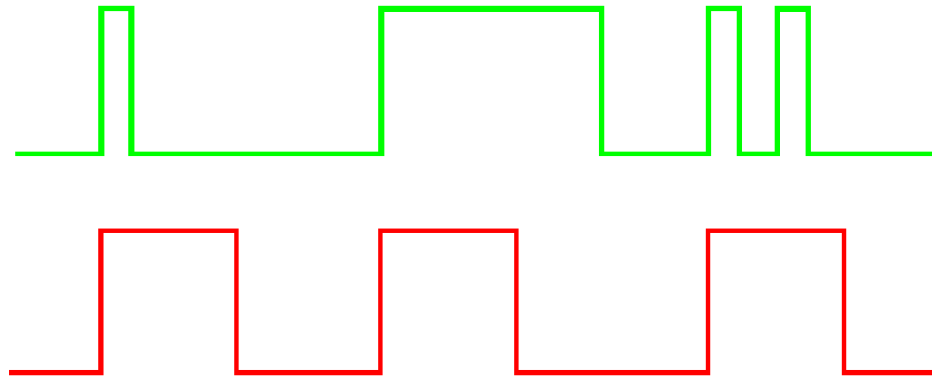
- Monostable circuits
- Astable circuits
- Schmitt Triggers

Regenerative circuits

- Bi-stable
 - Latches, flip-flops
 - Both output 0 and 1 can be maintained indefinitely as long as power is applied without any input signal.
- Monostable
 - Without any input, memory always returns to its one stable state after a period of time.
 - Receiving an input temporarily places circuit in its unstable output state.
 - Sometimes called a “one-shot”
- Astable
 - No stable state.
 - Memory oscillates between both output states at a fixed time interval.
- Sometimes, particularly in mono and astable configuration, regenerative circuits called “multivibrator”.
 - Ex: “Astable multivibrator”.

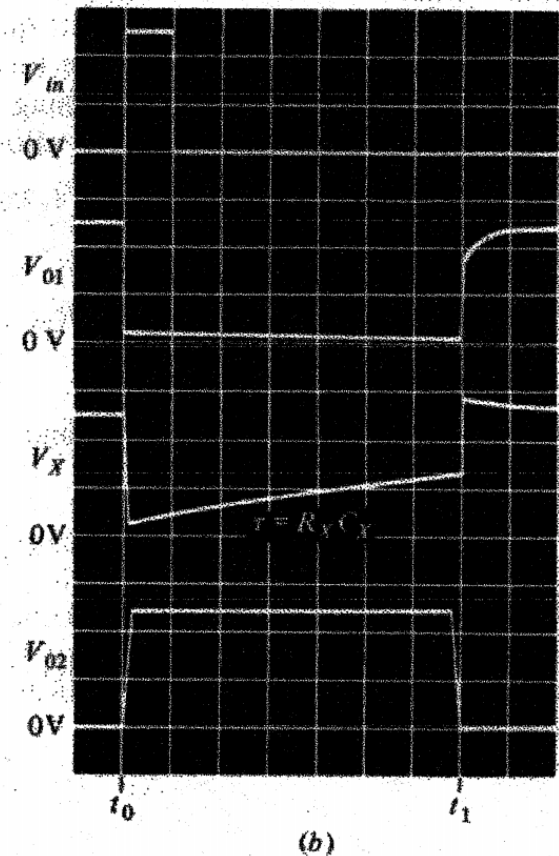
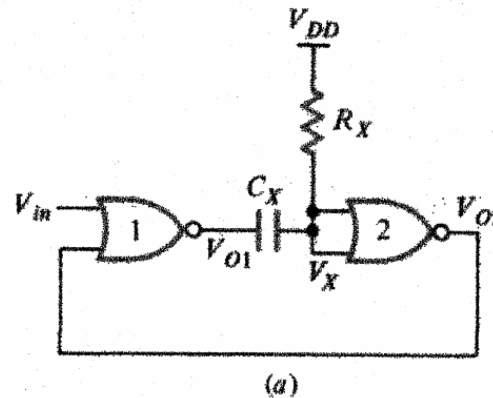
Monostable Circuits

- Applications
 - Pulse elongation.
 - Ensure that a low clock-speed detector can detect a short time period pulse.
 - Useful in Analog, mixed frequency signal detection.



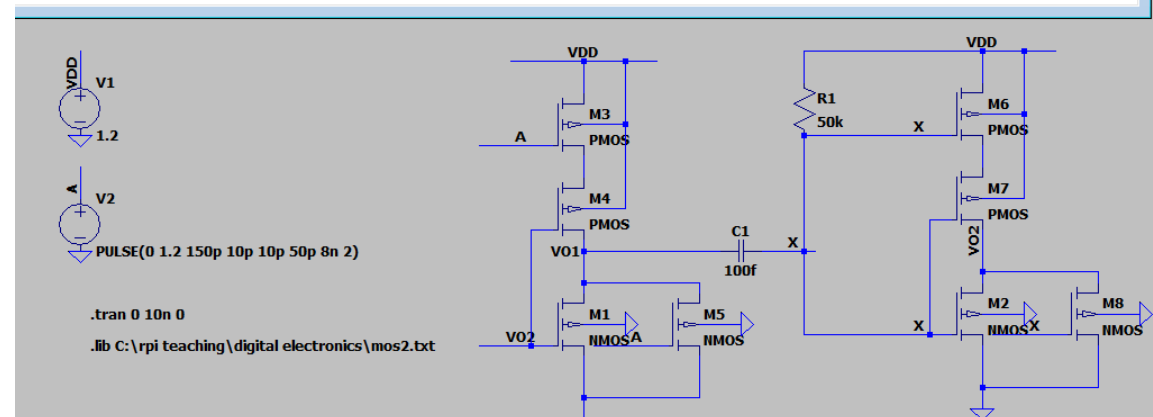
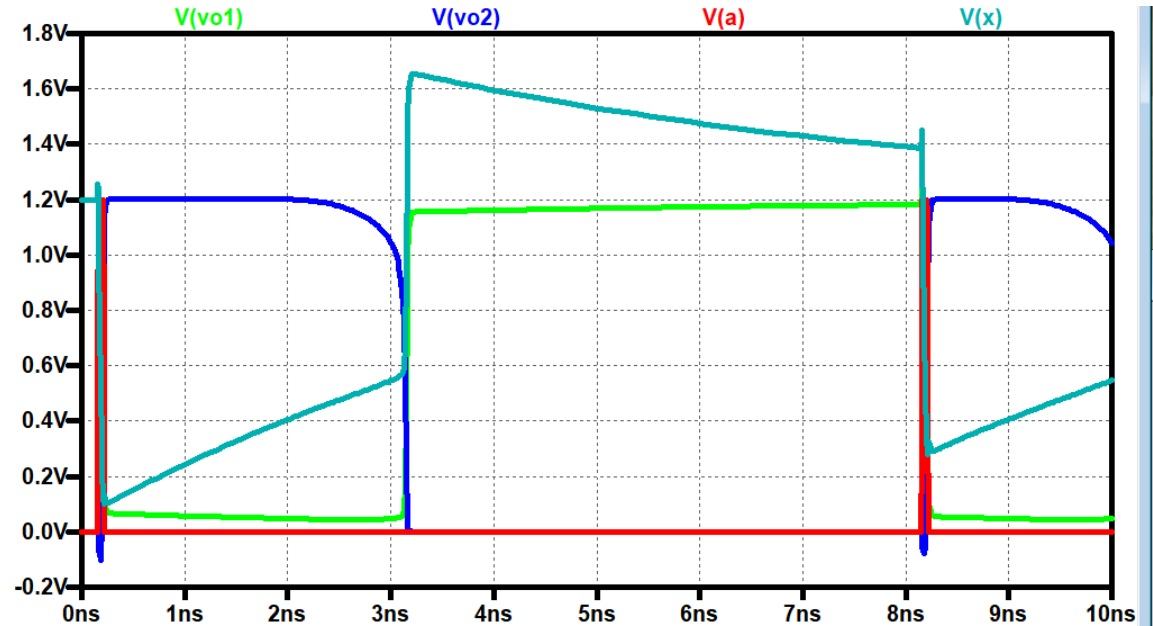
Monostable Circuit

- Series SR latches
- $\tau = 0.7 * R_X C_X$
- High speed V_{O1} output passes through C_X
 - C_X must be large enough pass to V_X , overpower R
 - Low enough to allow cir
- $NOR_1 R_{eff}$ overpowers I
 V_X low



Monostable Circuit Example

- $R_X = 50 \text{ k}\Omega$
- $C_X = 100 \text{ fF}$



τ_{hold}

- V_{O2} reverts to output 0 when V_X reaches V_S ($V_{DD}/2$)
- $\tau_p = 0.7RC$ is delay for 50% voltage swing, same as V_S
- $\tau_{\text{hold}} = 0.7 * R_X C_X$
 - In this example, $R_X = 50 \text{ k}\Omega$, $C_X = 100 \text{ fF}$
 - $\tau_{\text{hold}} = 0.7 * 50 \text{ k} * 100 \text{ f} = 3.5 \text{ ns}$

V_X Node Voltage

$$\text{NOR } \tau_{\text{fall}} \approx 10 \text{ ps}$$

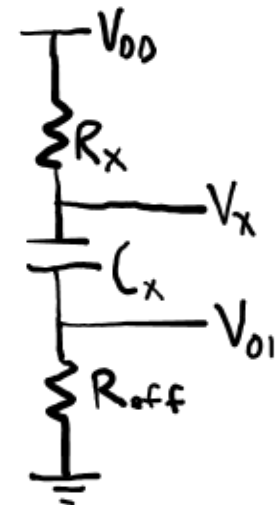
$$Z = \frac{1}{j\omega C}, \quad \omega = 2\pi f = \frac{2\pi}{\tau_{\text{rise}}} = \frac{2\pi}{10 \text{ p}} = 630 \text{ GHz}$$

$$C = C_X, \quad C_X = 100 \text{ fF in example}$$

$$Z = \frac{1}{j \cdot 630 \cdot 10^9 \cdot 100 \cdot 10^{-15}}$$

$$Z \approx 1.6 \text{ k}\Omega \quad Z_{C_X} + R_{\text{eff}} = 1.6 \text{ k} + 12.5 \text{ k} = 14.1 \text{ k}\Omega$$

$Z_{C_X} + R_{\text{eff}}$ is less than R_X (50k),
allowing V_X to be pulled low.

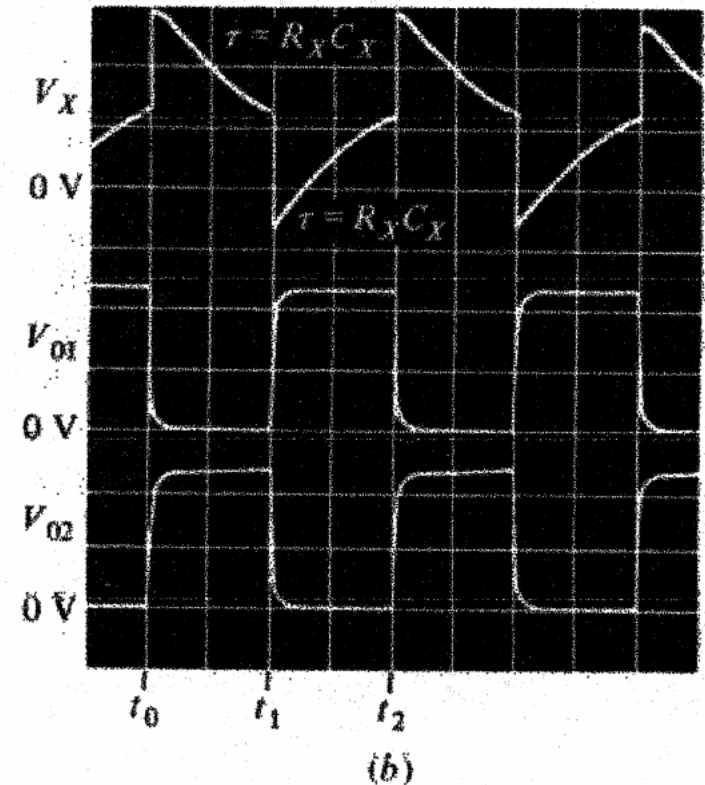
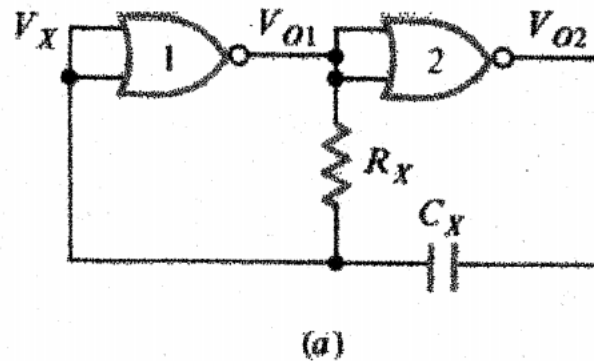


Astable Circuits

- Applications
 - Clock generation
 - Relaxation oscillator
 - Cheap!

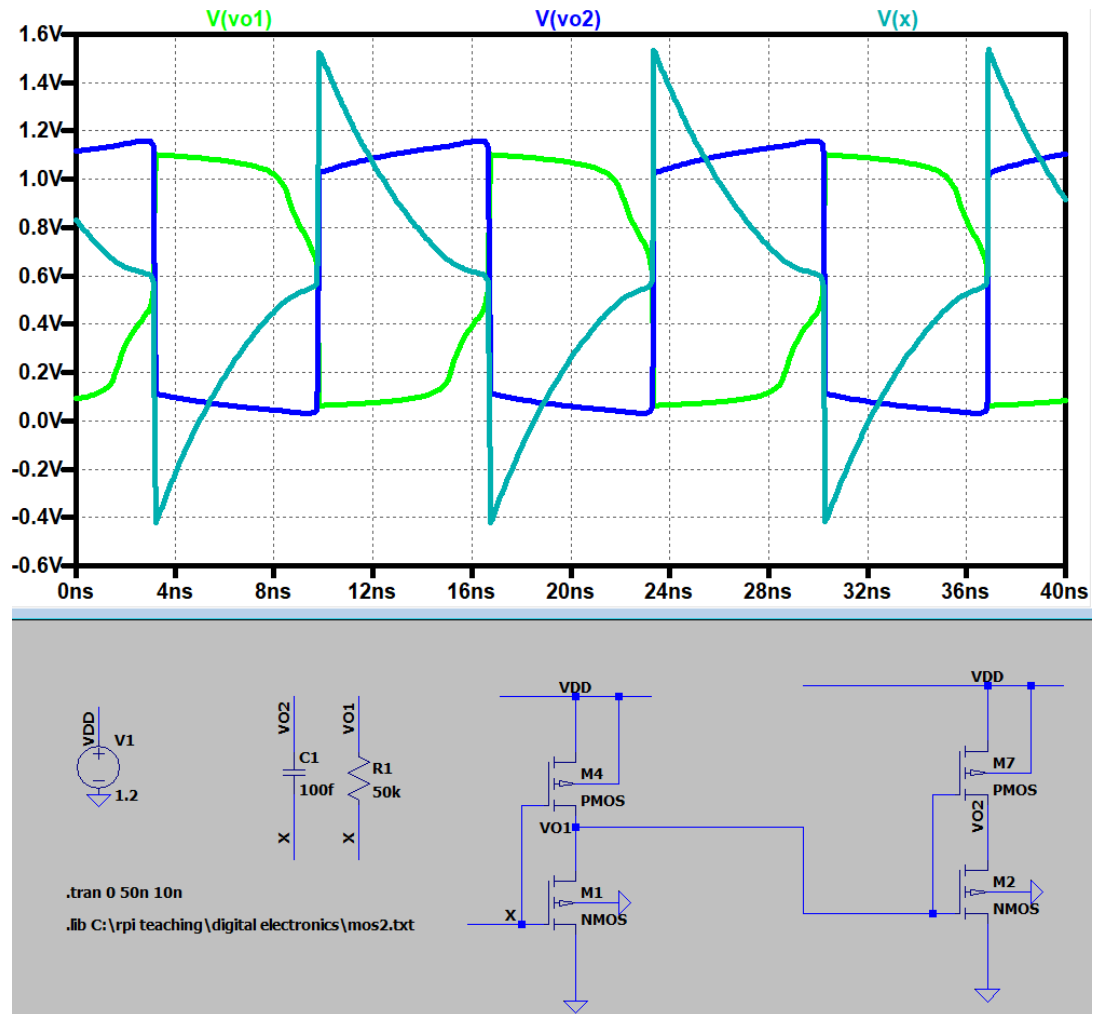
Astable Circuit

- Series SR latches (can be inverters)
- $\tau = 1.1 * R_X C_X$
- High speed V_{O2} output change passes through C_X
 - C_X must be large enough to allow V_{O1} to pass to V_X , overpower R_X .
 - Low enough to allow circuit to reset.
- $\text{NOR}_2 R_{\text{eff}}$ overpowers R_X , oscillating V_X
- Over time, R_X sets $V_X = V_{O1}$, switching all states.



Astable Circuit Example

- $R_X = 50 \text{ k}\Omega$
- $C_X = 100 \text{ fF}$



$$\tau_{\text{osc}}$$

- V_{O1} oscillates when V_X reaches V_S ($V_{DD}/2$)
- $\tau_p = 0.7RC$ is delay for 50% voltage swing, same as V_S
 - However, we are no longer waiting for V_X to settle to V_{DD} or GND.
 - For positive transition V_X voltage is charged to $\sim V_S + V_{DD}$, $1.5*V_{DD}$, or 1.8V in the worst-case scenario.
 - This results in a slower 100% to 33% voltage transition, where $\tau = 1.1*R_X C_X$
 - Issue is further weakened in this case, where R_{eff} is not much lower than R_X
- $\tau_{\text{osc}} = 1.1*R_X C_X$
 - In this example, $R_X = 50 \text{ k}\Omega$, $C_X = 100\text{fF}$
 - $\tau_{\text{osc}} = 1.1*50\text{k}*100\text{f} = 5.5 \text{ ns}$

V_X Node Voltage

NOR $\tau_{fall} \approx 10 \text{ ps}$

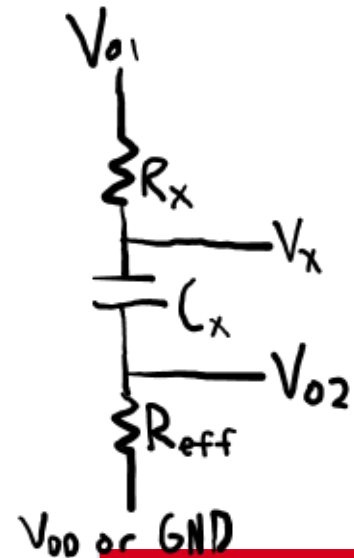
$$Z = \frac{1}{j\omega C}, \quad \omega = 2\pi f = \frac{2\pi}{\tau_{rise}} = \frac{2\pi}{10 \text{ p}} = 630 \text{ GHz}$$

$C = C_X, C_X = 100 \text{ fF}$ in example

$$Z = \frac{1}{j \cdot 630 \cdot 10^9 \cdot 100 \cdot 10^{-15}}$$

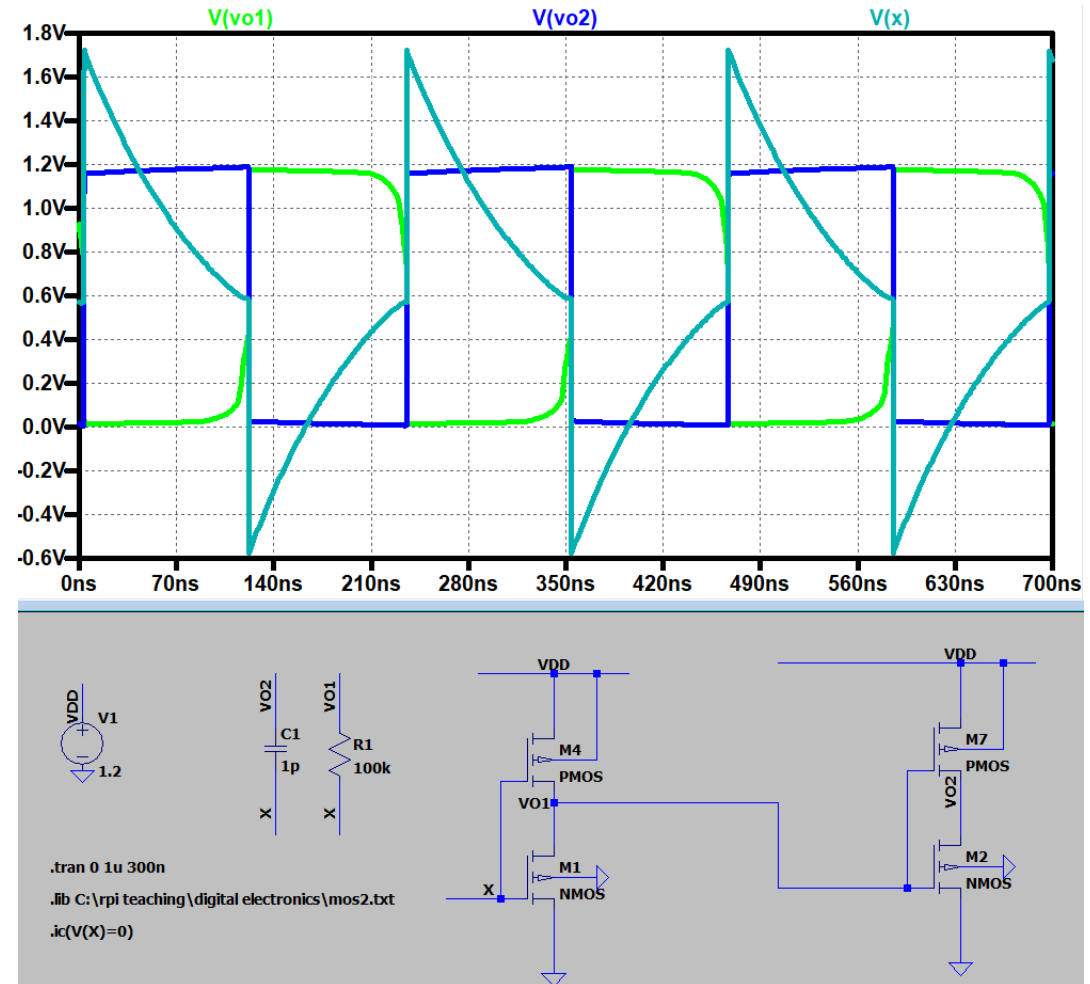
$$Z \approx 1.6 \text{ k}\Omega \quad Z_{CX} + R_{eff} = 1.6 \text{ k} + 12.5 \text{ k} = 14.1 \text{ k}\Omega$$

$Z_{CX} + R_{eff}$ is less than $R_X (50 \text{ k})$,
allowing V_X to be controlled by V_{O2} .



More Ideal Astable Circuit

- $R_X = 100 \text{ k}\Omega$
- $C_X = 1 \text{ pF}$
 - Lower cutoff frequency, less attenuation
- Widths doubled
 - Faster rise, fall to pass better through C_X
 - Lower R_{eff} to set V_{O1} closer to V_{DD} and GND.
- $\tau_{\text{osc}} = 1.1 * R_X C_X$
- $\tau_{\text{osc}} = 1.1 * 100\text{k} * 1\text{p}$
- $\tau_{\text{osc}} = 110 \text{ ns}$

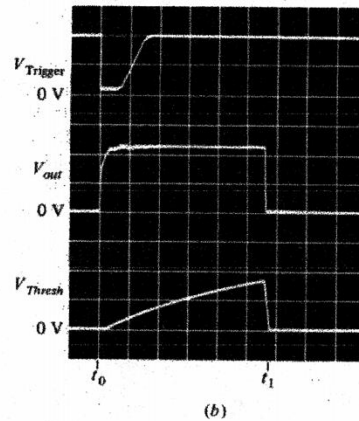
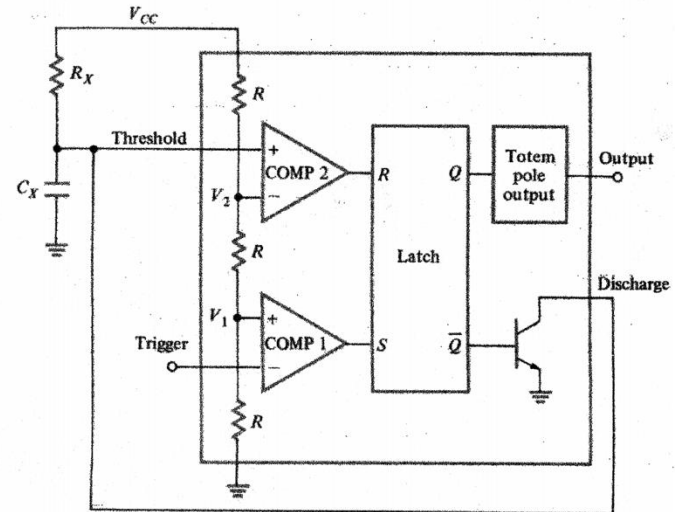


555 Timer

- Industry staple for decades.
- Still commonly used by hobbyists, prototyping.
- Can be configured as monostable or astable.
- Allows for varied pulse width, duty cycle, and frequency.

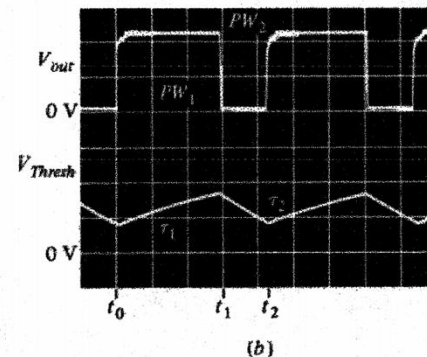
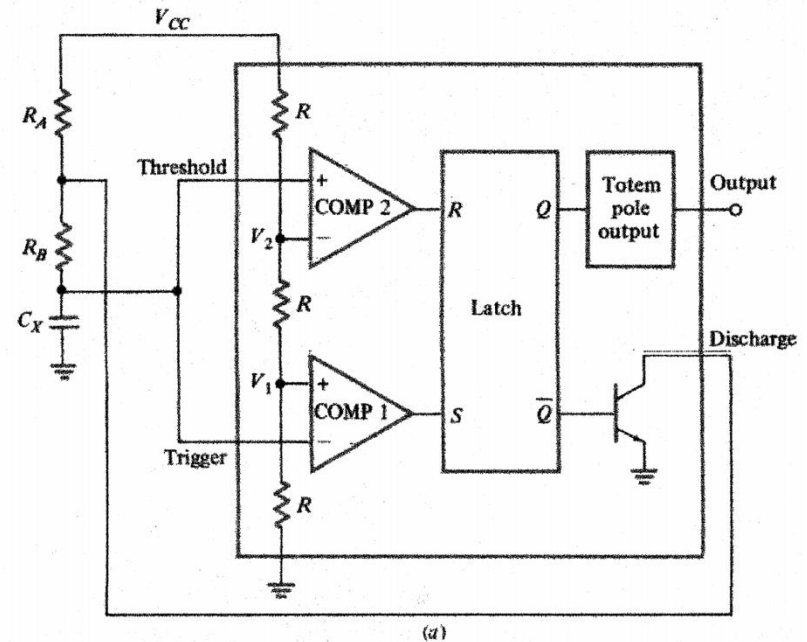
555 Timer

- Monostable configuration
- $\tau_{\text{hold}} = 1.1 * R_X C_X$
 - $V_{\text{threshold}}$ compared to $V_{\text{DD}} * 2/3$, not $V_{\text{DD}}/2$



555 Timer

- Astable configuration
- $\tau_{\text{high}} = 0.7 * (R_A + R_B) C_X$
- $\tau_{\text{low}} = 0.7 * R_B C_X$

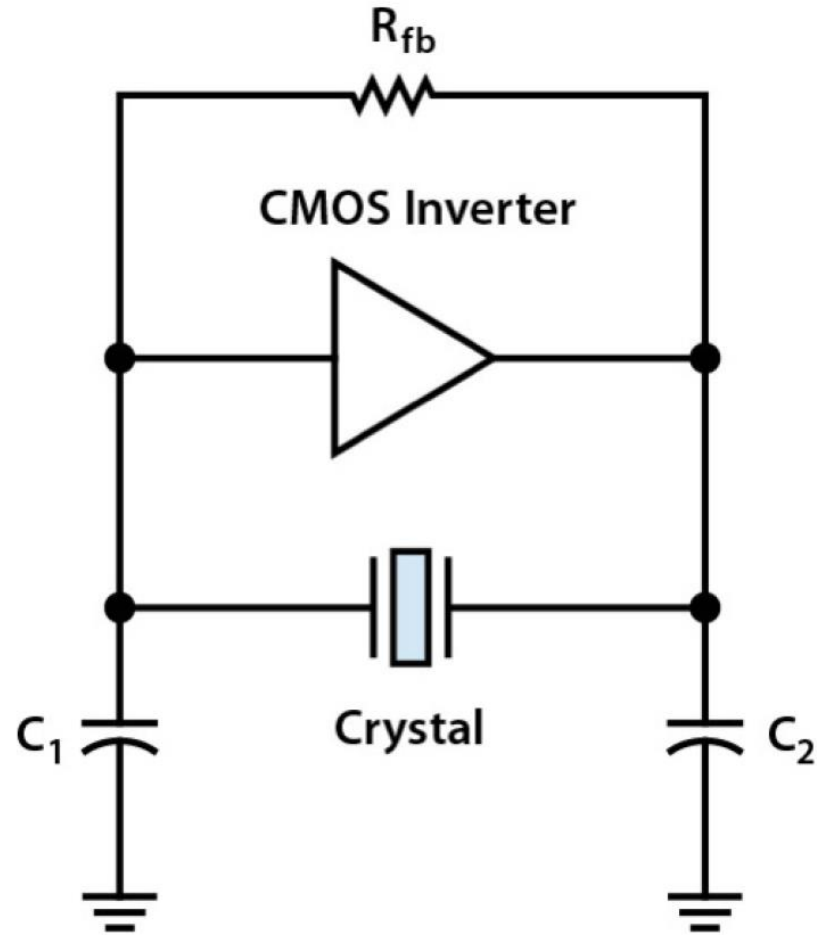


Crystalline Oscillator

- Use of relaxation oscillator becoming increasingly rare.
 - More demand for higher clock accuracy.
 - R, C values vary in manufacturing.
 - R, C values vary with temperature.
 - Modern ICs require better than 1-2% clock accuracy.
 - More accurate crystalline oscillators becoming cheaper.
 - Orders of magnitude superior clock accuracy.
 - Resistant to temperature change.

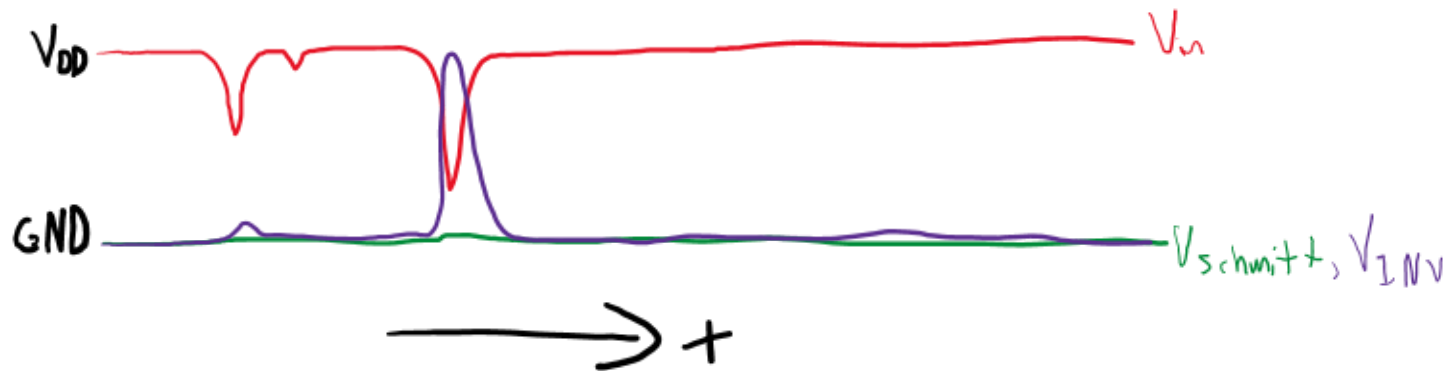
Crystalline Oscillator

- Reverse piezoelectric effect
 - Apply voltage to crystal, crystal flexes.
 - Remove applied voltage, crystal rebounds
 - Oscillates at designed fixed frequency.
- Pierce Oscillator common
 - INV restores crystalline resistive losses.
 - R_{fb} biases INV MOSFETs in saturation region for more gain
- Multiply, divide with Phase-Locked Loop (PLL).



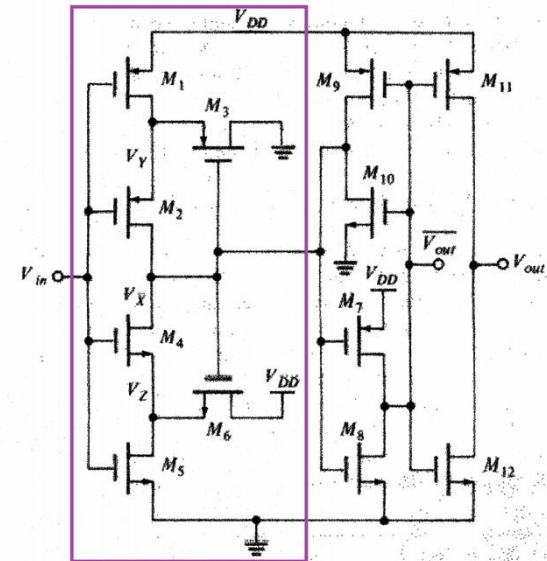
Schmitt Trigger

- In some applications, standard inverter has insufficient noise margins.
 - Noise margin cannot exceed $V_{DD}/2$
- Passive RC filter degrades signal.
- Schmitt Trigger provides means of regenerative noise filtering.

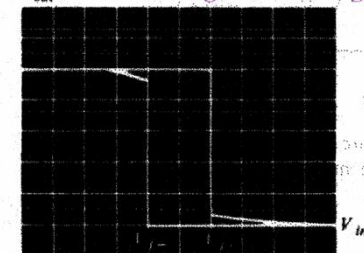


Schmitt Trigger

- Basic building block -
Hysteretic Inverter
 - Inverter with different V_S depending on state
 - High V_S if output low
 - Low V_S if output high.

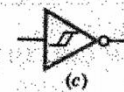


(a) One inverting Schmitt trigger



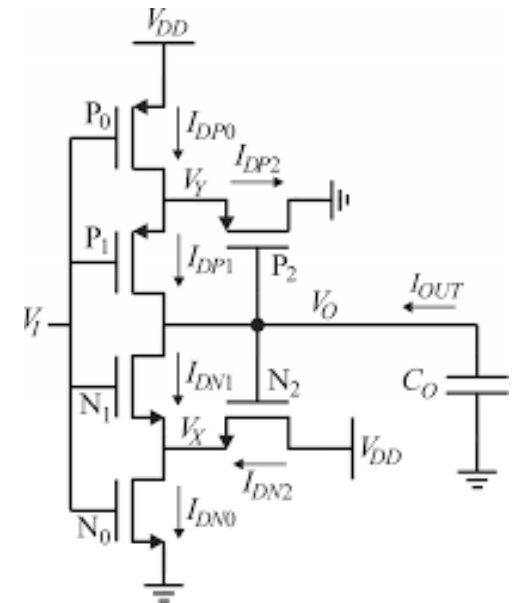
$X = V_{in}$ (1 V/div)
 $Y = V_{out}$ (2 V/div)

(b)



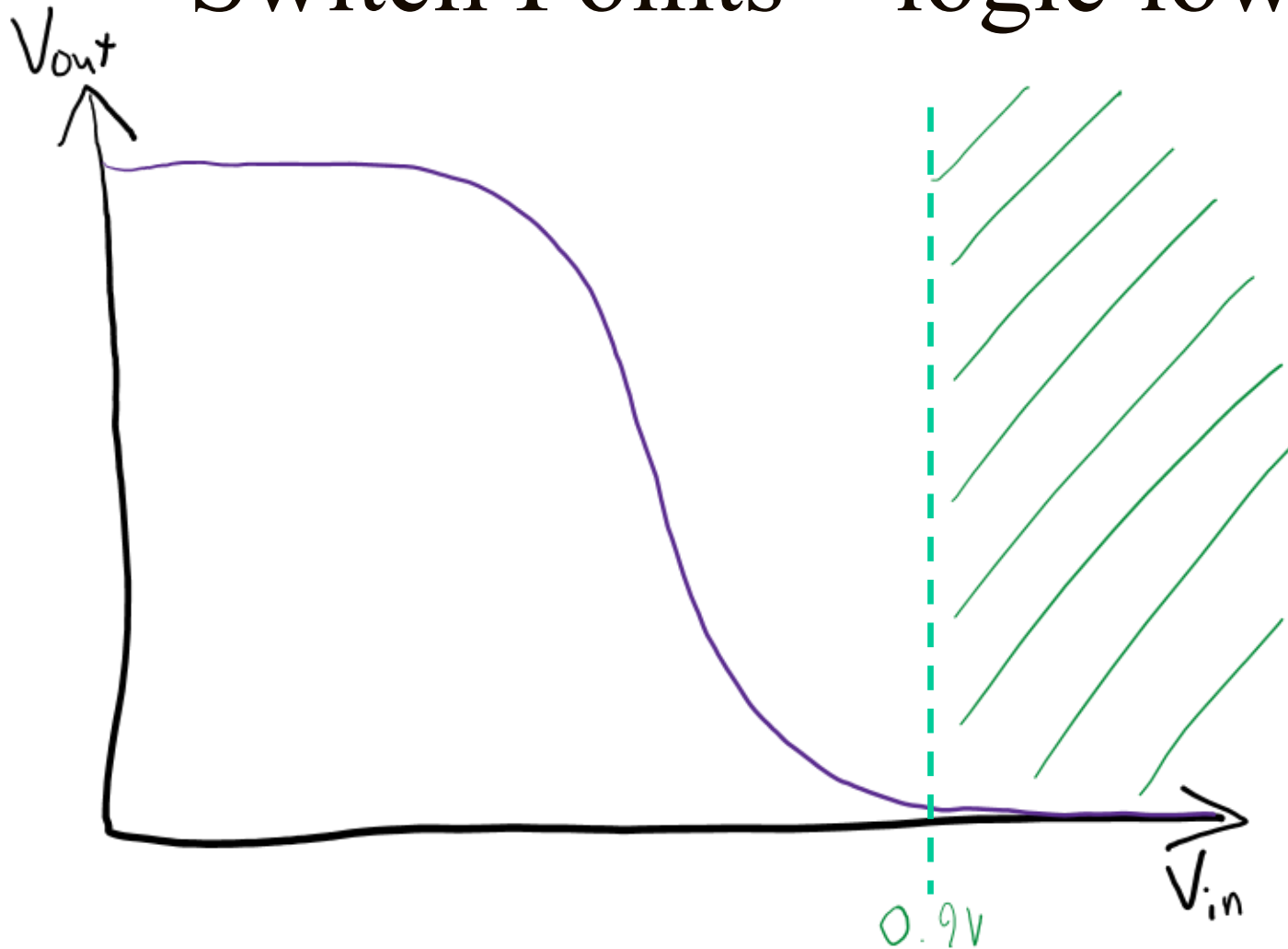
Switch Points

- Think of P1, N1 as the core inverter.
 - P0 and N0 conditionally cause shift in V_{source} .
 - Increase GND voltage when output high.
 - Decrease V_{DD} voltage when output low.
- N2, P2 weak compared to INV totem.
- V_X node set by N0, N2 inverter with active NMOS N2 pull-up.
 - When V_{OUT} is high, steady-state current. $I_{DN0} = I_{DN2}$.
 - $V_X \sim 0.3V$
 - When V_{OUT} is low, N2 is off, $V_X = 0V$
- V_Y node set by P0, P2 inverter with active PMOS P2 pull-down.
 - When V_{OUT} is low, steady-state current. $I_{DP0} = I_{DP2}$.
 - $V_Y \sim 0.9V$
 - When V_{OUT} is high, P2 is off, $V_Y = 1.2V$



<https://ieeexplore.ieee.org/document/7795159>

Switch Points – logic low



Switch Points – logic high

